

Inverse Folding Models Sample Sequence Space Shaped by MHC Anchor Identity and TCR Context

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Abstract

Inverse-folding models have emerged as potent tools for designing protein sequence from structure, and the degree to which they capture the biophysical constraints of a specific molecular interface stands as an open question. TCR cross-reactivity has caused fatal toxicity in the clinic, yet the peptide sequence space is far too large to screen experimentally. We therefore ask whether inverse-folding models can recover the native peptide from structure alone, and what their sampling distribution reveals about the constraints that give rise to cross-reactivity. We applied four inverse-folding models to twenty solved pMHC-TCR structures under two structural contexts, generating 1.6 million peptide designs, scored where allele-matched against two independent binding predictors, and characterized the full sampling distribution each model induces over peptide sequence space. No model recovered the native peptide. Recovery is instead bimodal across the two anchor positions that dominate MHC binding. The C-terminal anchor P Ω , though structurally load-bearing, is largely missed, while the position-2 anchor recovers at or above the interior average. Removing the T-cell receptor increases sampling diversity across all four models and raises predicted binding affinity in a model-dependent manner, but lowers recovery at experimentally validated contact positions and makes every model that reports a confidence score less confident in its own designs, showing that the models read the receptor as part of the interface. This suggests that inverse-folding models encode peptide compatibility as a set of position-specific structural constraints rather than a single optimal sequence. Our investigation establishes the sampling distribution as a probe of the constraints governing peptide recognition, and highlights both the promise and the limitations of structure-based models for predicting cross-reactivity.

1 Introduction

T-cell receptor (TCR) engineering therapies promise a transformative future for cancer treatment, and the first reached approval in 2024 [7, 12], yet cross-reactivity, the recognition of unintended peptide-MHC (pMHC) complexes, remains a fundamental barrier to scaling them into personalized medicine [20, 31]. An affinity-enhanced MAGE-A3 receptor that had cleared extensive preclinical testing ulti-

mately caused fatal cardiac toxicity in two patients, cross-recognizing an unrelated pMHC derived from titin [5, 17].

Experimental methods therefore sample only a fraction of the pMHC-TCR interaction space, which is combinatorially vast, roughly 20^L for a peptide of length L , yet far from uniform. The anchor sites collapse the entropy at fixed positions while leaving the remainder variable [11, 23], so the effective dimensionality of the presented distribution falls far below its combinatorial ceiling. It is this low-dimensional structure that makes the problem learnable, yet the same data limitations that bound experimental sampling also limit the models trained on that data [22, 24], motivating approaches that exploit structural rather than purely sequence-based signals.

Inverse-folding models sample sequences from a distribution conditioned on structure [8, 13], and on the receptor side of this interface, the approach has recent precedent. Comparing ProteinMPNN and ESM-IF1 against physics-based Rosetta design [1] across 38 solved TCR-pMHC complexes, prior work [25] found that both inverse-folding models recover native CDR3 interface residues well above chance, establishing that inverse folding captures real, non-trivial signals from a TCR-pMHC interface.

Here, we turn the same class of models toward the peptide itself. The point estimate asks whether the model recovers the native peptide the structure was solved with; the distribution asks which other peptides the model treats as compatible with the same groove and the same receptor, precisely the variant and off-target sequences that underlie cross-reactivity. Because inverse-folding models score structural compatibility rather than a labeled binding phenotype, that distribution provides a first-pass map of the tolerated peptide neighborhood around the native ligand, and, because MHC-groove geometry and TCR-contact geometry occupy structurally distinguishable positions along the peptide, acts as a probe of which positions the model treats as side-chain-constrained versus permissive. We therefore ask whether inverse-folding models recover the native peptide, and whether the distribution they induce over sequence space concentrates on the tolerated neighborhood underlying cross-reactivity and off-target recognition.

2 Materials and Methods

2.1 Panel and models

Twenty pMHC-TCR crystal structures were retrieved from the Protein Data Bank [2] (Supplementary Table S1). Every structure includes a bound peptide, the MHC class I heavy chain, β_2 -microglobulin, and a complete TCR $\alpha\beta$ heterodimer [3] resolved in the same asymmetric unit with peptides of various lengths (Supplementary Figure S1a). 19 structures are HLA-A*02:01; the remaining structure, 4MJI, is HLA-B*51:01 and was excluded from the allele-specific binding predictor. The average crystallographic resolution is 2.46 (Supplementary Figure S1b).

The four models are ProteinMPNN [8], a message-passing GNN encoding the fixed backbone as a k -nearest-neighbor graph and decoding sequence autoregressively; ProteinMPNN (no MHC), the same architecture retrained with MHC/TCR-class structures excluded; ESM-IF1 [13] and LigandMPNN [9] (Supplementary Table S2).

Each of the four tools samples only the peptide chain, with the MHC, β_2 -microglobulin, and (where present) TCR chains held fixed as structural context. Two context conditions were run for every structure, MHC + β_2 m + TCR $\alpha\beta$ and the TCR removed from the input structure entirely. Each condition was sampled to 10,000 raw designs at a fixed sampling temperature of $T = 0.1$.

3 Results

3.1 Models Sample Strong Position-Specific Sequence Bias

To establish how much of the peptide sequence space these models actually explore, we sampled 10,000 designs per (structure, model) cell in the pMHC+TCR context and counted how many were distinct. Only 50 designs per cell were unique on average, so these models return redundancy rather than diversity (Supplementary Figure S2). Sequence logos [27] show every model imposes strong position-specific restriction independent of recovery (Supplementary Figures S3–S4d), a property of the sampling distribution rather than evolutionary conservation, tracing the same contacts seen in the crystals, where 23 of 40 MHC positions contacting the peptide anywhere in the panel do so in all twenty (Supplementary Figure S1c).

3.2 The two MHC anchors impose different constraints and recover differently

To determine which peptide positions carry recoverable signal, we measured per-position recovery of each crystal’s own native residue, pooled across twenty structures and four models. Recovery is bimodal rather than uniform, and the two positions where this matters most are the two positions immunology already identifies as anchors [11, 18] (Figure 1). P2, seated in pocket B, constrains a specific side chain and recovers well above the interior average (0.64 vs. 0.49), whereas P Ω , seated in pocket F, imposes a permissive backbone constraint and recovers significantly *below* it (0.19 vs. 0.49, $p = 0.0054$). Both are structurally indispensable, yet only P2 gives a design model a signal it can exploit.

To ask whether that asymmetry reflects side-chain identity alone or a coarser physicochemical preference, we compared each native anchor residue against its own peptide’s interior positions on the Kyte-Doolittle hydrophobicity scale [16], and separately scored every design for whether it preserved the native residue’s side-chain class (Supplementary Figure S12). Both anchors are markedly more hydrophobic than the interior they sit alongside (3.44 at P2 and 3.65 at P Ω , against an interior mean of 0.79; $p = 0.0013$ and $p = 0.0001$, paired Wilcoxon across 19 structures), confirming that both pockets recruit hydrophobic residues. The two anchors nonetheless part company on how tightly that preference is enforced in the designs. P2 designs preserve the native side-chain class well above the interior baseline (85.6% vs. 70.7%, $p = 0.019$), whereas P Ω designs do not (62.4% vs. 70.7%, $p = n.s.$), even though P Ω designs do show significantly reduced hydrophobicity variance relative to the interior ($p = 0.041$). P Ω therefore behaves as a pocket that biases toward a hydrophobic character without fixing which hydrophobic residue satisfies it, which is what a low exact-identity recovery at a structurally load-bearing position should look like.

Averaged across the entire peptide, the four models fall within a narrow band of mean recovery (0.41 to 0.47 in full context and 0.28 to 0.36 without the TCR; Supplementary Figure S5), narrow enough that the models might appear indistinguishable by inspection. A Friedman test indicates otherwise in both contexts ($\chi^2 = 10.80$, $p = 0.013$, full context; $\chi^2 = 12.18$, $p = 0.007$, mhonly), so this apparent similarity reflects proximity in magnitude rather than statistical equivalence. As an aside, on 3QDG the central Gly-Ile-Gly motif at positions 4–6, well characterized in the literature [4], recovers essentially perfectly here (1.00, 0.73, 1.00 vs. 0.49), though we have not attempted a systematic motif annotation across the full panel.

The relationship between resolution and recovery illustrates how pooling across a heterogeneous panel can understate a genuine effect. Pooled across all twenty distinct peptides, the correlation is modest, and only one model survives correction individually ($r = -0.49$ pooled, $p = 0.030$; ESM-IF1 alone, $r = -0.63$, $p = 0.003$; Supplementary Figure S6). However, four of the twenty structures (2P5W, 2BNR, 2F53, and 2F54) are independent crystals of the identical peptide (SLLMWITQC, the NY-ESO-1 epitope [6, 26]) differing primarily in resolution (1.9–2.7 Å). Holding peptide identity fixed in this way removes a confound that the pooled panel cannot, and the effect becomes unambiguous. Declining resolution drives recovery down for two of the four models ($r = -0.99$, $p = 0.012$, ProteinMPNN (no MHC); $r = -0.97$, $p = 0.031$, ESM-IF1) and drives unique design counts up for the remaining two ($r = 0.97$, $p = 0.028$, ProteinMPNN; $r = 0.96$, $p = 0.040$, ProteinMPNN (no MHC); Supplementary Figure S7). The pooled-panel estimate was therefore not incorrect, but diluted by heterogeneity.

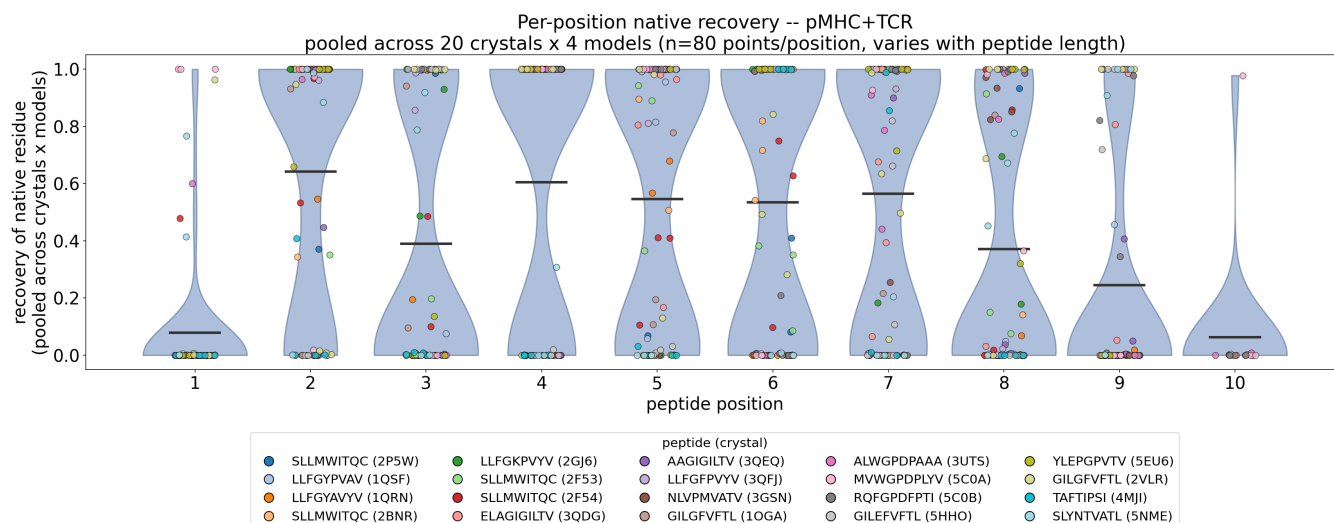


Figure 1: **Per-position recovery is bimodal, and the two MHC anchors diverge.** Recovery of the native residue, pooled across all 20 crystals and 4 models, one violin per peptide position (pMHC+TCR context). Individual points are colored by crystal.

3.3 Most peptides sampled by inverse-folding models are predicted to be presented

Having established the recovery rates of the native peptide and the sequence space each structure samples, we next asked whether those sampled peptides would actually be presented, scoring every unique design from the nineteen HLA-A*02:01 structures for predicted HLA-A2 binding with two unrelated predictors, MHCflurry [22] and ESM-CBA [10, 19], neither of which had seen this panel (Figure 2).

The two predictors agree closely on the same designs ($r = 0.83$ – 0.91 per model in full context, up to 0.94 without the TCR), and agreement between two methodologically unrelated predictors constitutes stronger evidence than either predictor alone. On this shared landscape, projected jointly into one UMAP embedding space [21] (Supplementary Figure S11), ProteinMPNN and LigandMPNN consistently sample toward higher predicted affinity than ESM-IF1. At the 500 nM cutoff [28], 61 – 78% of ProteinMPNN-family and LigandMPNN designs qualify by MHCflurry, compared with 45% for ESM-IF1 ($\chi^2 = 238.46$, $p = 2 \times 10^{-51}$; see also Section 3.4). This pattern is better characterized as ESM-IF1 being a consistent low-affinity outlier than as a general superiority of ProteinMPNN over ESM-IF1, since the other three tools do not share this deficit.

3.4 Without the TCR, inverse-folding models sample more widely but less confidently

To understand the effect of the TCR on what these models propose, we resampled every structure with the TCR deleted from the input and compared the resulting designs against the full-complex designs above.

Every model explores substantially more sequence space without the TCR, with unique design counts roughly doubling or tripling (ProteinMPNN $1081 \rightarrow 3671$; Protein-

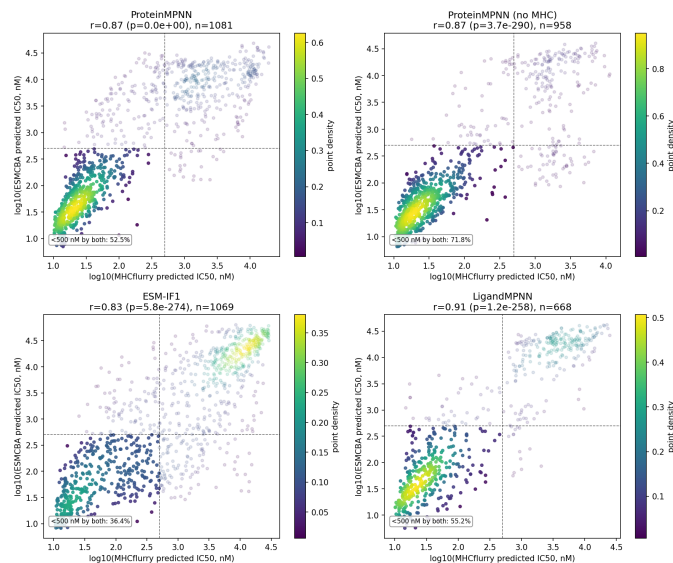


Figure 2: **MHCflurry and ESM-CBA predictions of designs** $\log_{10}(\text{predicted IC}_{50}, \text{nM})$ with pMHC+TCR context, one panel per model. Dashed lines mark the 500 nM ; The percentage is the fraction of designs under 500 nM by both predictors at once.

MPNN (no MHC) $958 \rightarrow 2362$; ESM-IF1 $1069 \rightarrow 1865$; LigandMPNN $668 \rightarrow 2111$; Supplementary Figure S5). This larger space skews toward higher predicted affinity, but not uniformly across models. ProteinMPNN (no MHC) shows the largest shift (strong-binder rate $77.5\% \rightarrow 95.2\%$), LigandMPNN a smaller but genuine one ($61.1\% \rightarrow 70.7\%$), while ProteinMPNN and ESM-IF1 show little change (Supplementary Figure S10). The affinity gain associated with

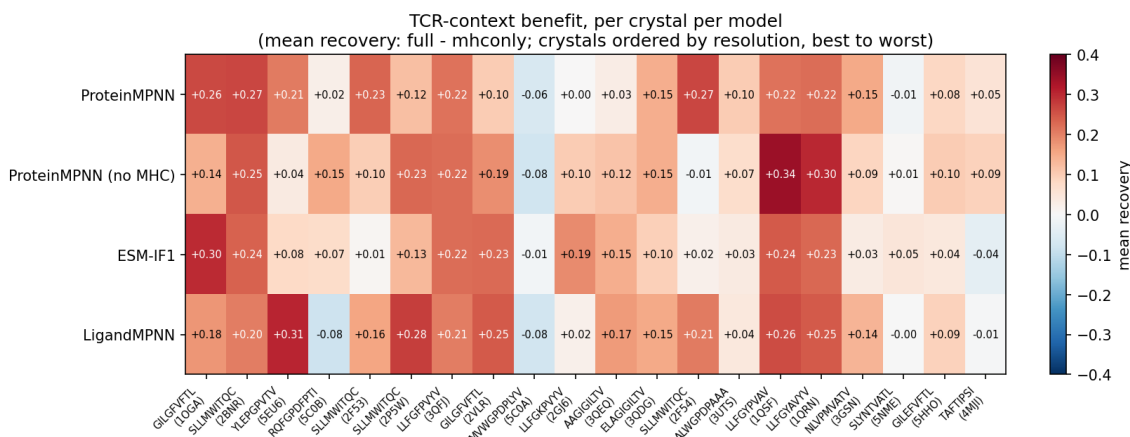


Figure 3: **Where the TCR-context benefit actually concentrates.** Recovery(full) – recovery(mhonly), per crystal per model, crystals ordered by resolution (best to worst, left to right).

TCR removal thus appears concentrated in the two models already most context-sensitive, rather than reflecting a general consequence of removing the TCR.

A related but distinct signal comes from each model’s own confidence in the sequences it designs, independent of either external predictor. ProteinMPNN and ProteinMPNN (no MHC) report an average per-residue negative log-likelihood for each design, for which lower values indicate higher confidence; LigandMPNN reports an overall confidence score, for which higher values indicate higher confidence. Pooled across all nineteen HLA-A*02:01 structures, removing the TCR shifts this internal score for every model, and in every case toward lower self-assessed confidence (ProteinMPNN mean 1.016 $\rightarrow$ 1.295; ProteinMPNN (no MHC) 0.925 $\rightarrow$ 1.128; LigandMPNN 0.413 $\rightarrow$ 0.321; Mann-Whitney $p < 10^{-200}$ for all three; Supplementary Figure S9a). This shift is not uniform across the panel. Broken out per structure and Bonferroni-corrected, it reaches significance in 15–17 of 19 structures depending on the model, but not in the remaining two to four (Supplementary Figure S9b). That a model can become less confident in its own designs by its own internal metric while simultaneously being scored as a stronger binder by an external predictor that never saw those designs indicates that the two signals are not redundant, since one reflects how comfortably a sequence fits the model’s learned distribution, the other how likely it is to bind, and removing the TCR does not move them the same way. This distinction is more informative than amino-acid-level recovery or composition alone, since it speaks to what the model itself, not just an external predictor, “believes” about the sequences it is generating.

3.5 Recovery Loss Concentrates at Experimentally Validated TCR-Contact Positions

To understand how these residue interactions contribute functionally to TCR binding, we tested whether the recovery lost without the TCR concentrates at positions that experi-

mental data already implicate in receptor engagement. We cross-referenced SKEMPI 2.0 [14] and IEDB [30] against each structure’s crystallized TCR clone (matched by CDR3 sequence) and identified 21 (structure, position) pairs with an experimentally measured mutation that alters TCR binding. Pooling recovery across exactly these positions, TCR-present designs recover the native residue 54–58% of the time, compared with 34–38% for TCR-absent designs (Figure 3), a consistent, though not universal, gap; a minority of individual positions move in the opposite direction.

4 Discussion

Inverse-folding models have rapidly advanced structure-based protein design [8, 9, 13], and their ability to recover native interface residues above chance [25] invites the question of whether they capture the biophysics of an interface or merely reproduce its most constrained positions. High recovery is easily read as the former. Our results point to the latter, since knowing that MHC anchor positions matter explains little on its own. P2 and P Ω are both structurally indispensable yet behave oppositely under inverse folding, one a side-chain readout, the other a backbone constraint tolerant of many side chains. This finer distinction is what a sequence-design model can meaningfully report. The same caution against pooling applies to resolution, where a weak correlation across twenty distinct peptides conceals a model-specific effect once peptide identity is controlled for, using the four SLLMWITQC/NY-ESO-1 crystals as a same-peptide replicate set. TCR removal resists a single verdict for the same reason. Diversity increases for every model, affinity for two, and recovery decreases on average but not everywhere, because whether the model requires the TCR was never one question.

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A Supplementary Figures

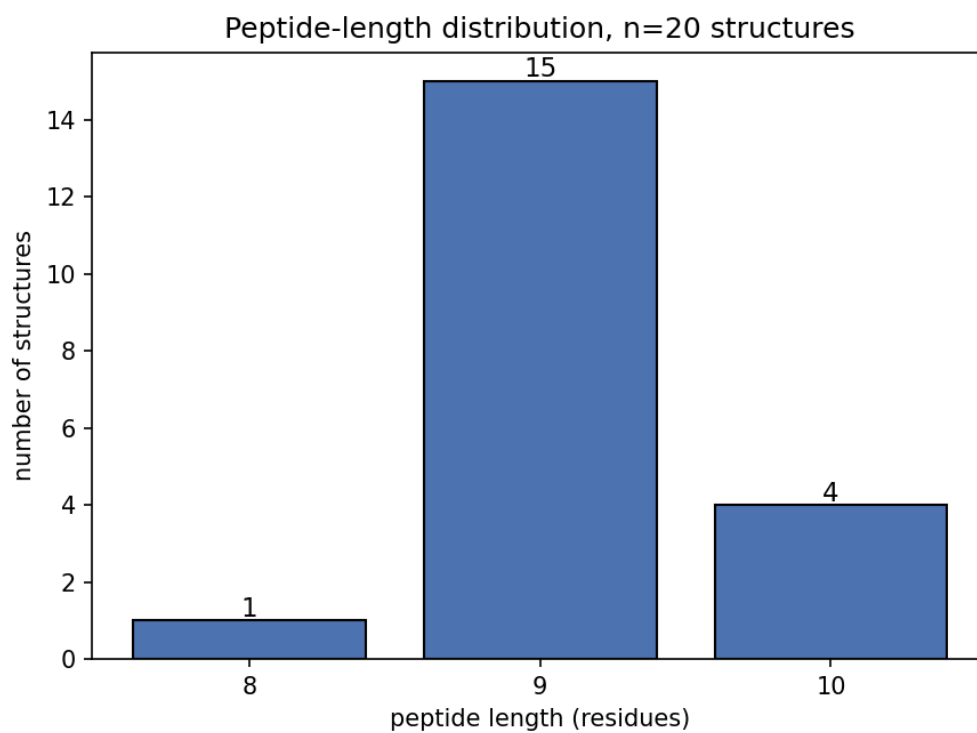


Figure S1a: **Peptide length distribution** across all 20 structures in the panel.

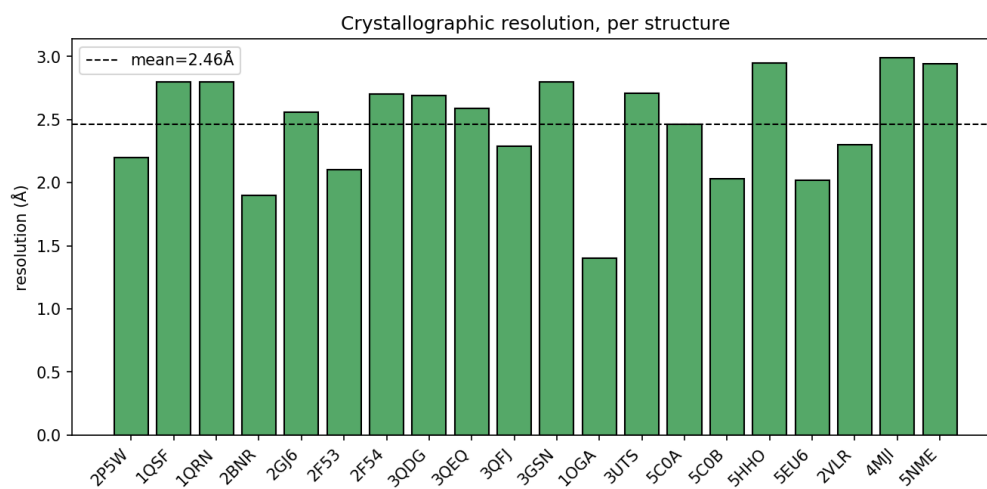


Figure S1b: **Crystallographic resolution per structure.**

PDB	Peptide	Length	Allele	Resolution (Å)
2P5W	SLLMWITQC	9	HLA-A*02:01	2.20
1QSF	LLFGYPVAV	9	HLA-A*02:01	2.80
1QRN	LLFGYAVYV	9	HLA-A*02:01	2.80
2BNR	SLLMWITQC	9	HLA-A*02:01	1.90
2GJ6	LLFGKPVYV	9	HLA-A*02:01	2.56
2F53	SLLMWITQC	9	HLA-A*02:01	2.10
2F54	SLLMWITQC	9	HLA-A*02:01	2.70
3QDG	ELAGIGILTV	10	HLA-A*02:01	2.69
3QEQ	AAGIGILTV	9	HLA-A*02:01	2.59
3QFJ	LLFGFPVYV	9	HLA-A*02:01	2.29
3GSN	NLVPMVATV	9	HLA-A*02:01	2.80
1OGA	GILGFVFTL	9	HLA-A*02:01	1.40
3UTS	ALWGPDPAAA	10	HLA-A*02:01	2.71
5C0A	MVWGPDPLYV	10	HLA-A*02:01	2.46
5C0B	RQFGPDFPTI	10	HLA-A*02:01	2.03
5HHO	GILEFVFTL	9	HLA-A*02:01	2.95
5EU6	YLEPGPVTV	9	HLA-A*02:01	2.02
2VLR	GILGFVFTL	9	HLA-A*02:01	2.30
4MJI	TAFTIPSI	8	HLA-B*51:01	2.99
5NME	SLYNTVATL	9	HLA-A*02:01	2.94

Table S1: **The twenty-structure panel.** PDB accession, native peptide, peptide length, MHC allele, and crystallographic resolution for every structure in the panel.

Model	Params	Architecture	Training data	Predicted structures
ProteinMPNN	1.66M	Message-passing GNN	PDB (pdb_2021aug02), general backbones	–
ProteinMPNN (no MHC)	1.66M	Message-passing GNN (identical to above)	Same corpus, MHC/TCR-class structures excluded	–
ESM-IF1	141.7M	GVP-GNN + transformer	CATH-derived backbones	+12M structures AF2
LigandMPNN	2.62M	Message-passing GNN + ligand context	PDB, augmented with explicit ligand/heteroatom context	–

Table S2: **Models compared.** Parameter counts verified directly from each checkpoint’s `state_dict`. ESM-IF1 is the only tool whose training set is built substantially from AlphaFold2-predicted structures [15] rather than solved crystal structures (~12M predicted backbones, on top of its CATH-derived crystal set [29]).

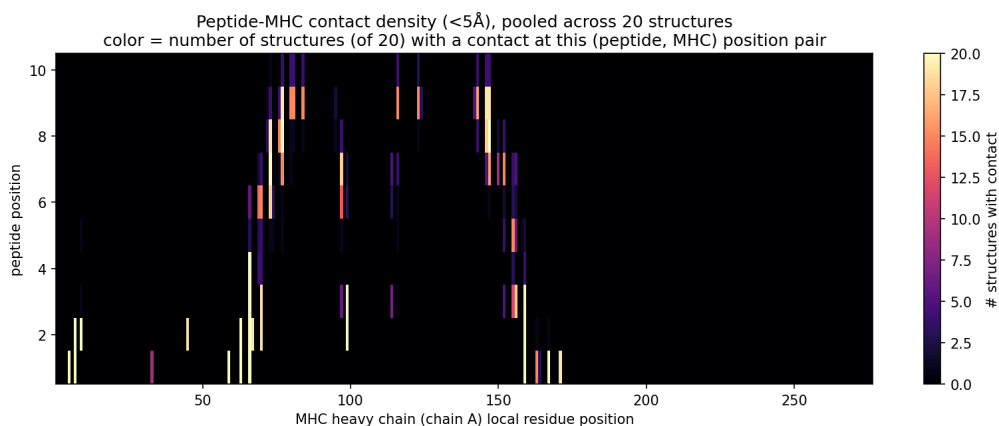


Figure S1c: **MHC residue vs. peptide contact density**, pooled across all 20 structures.

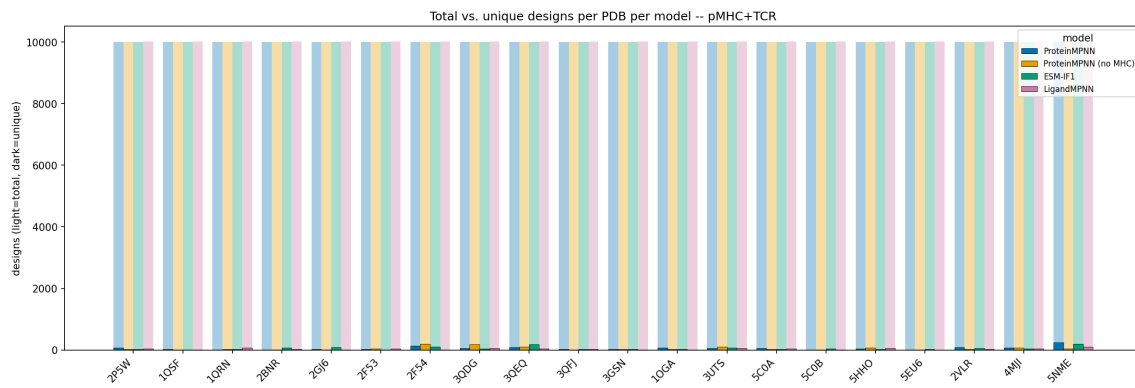


Figure S2: **Extreme redundancy at $T=0.1$.** 10,000 raw draws per (structure, model); the unique fraction is 0.50% on average and never exceeds a few percent.

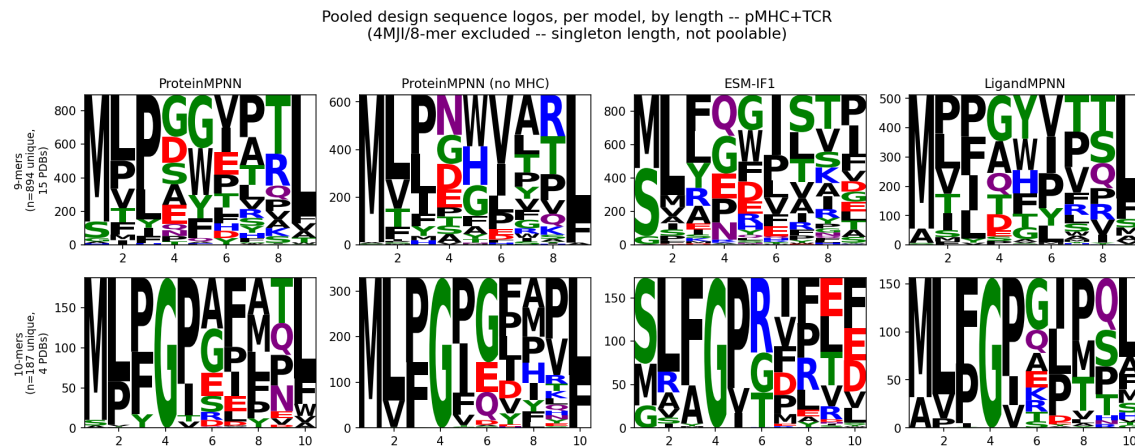


Figure S3: Design sequence logos, pooled by peptide length, one column per model.

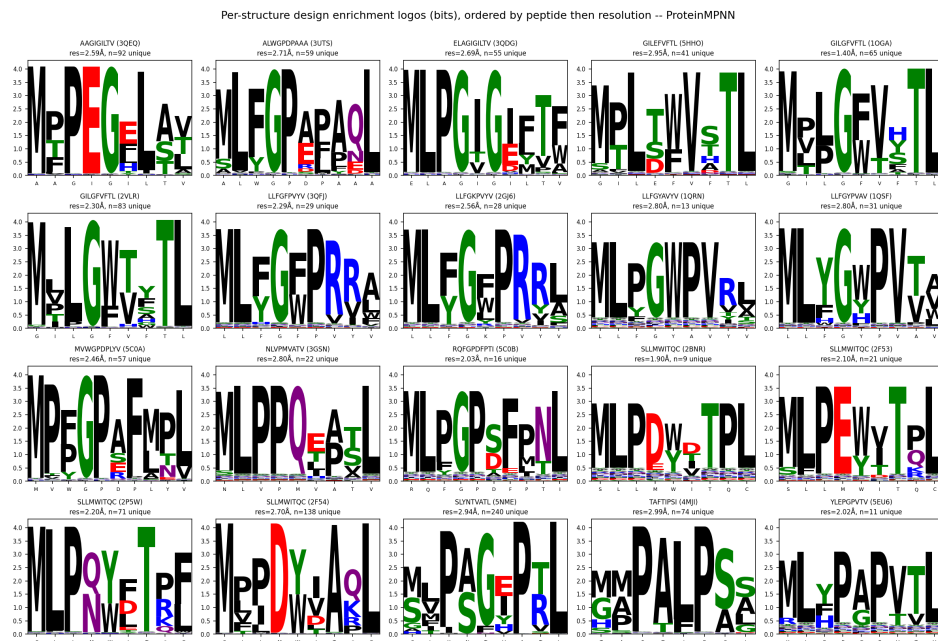


Figure S4a: **Per-structure design logos, ProteinMPNN.** All 20 structures, ordered by native peptide then resolution.

Per-structure design enrichment logos (bits), ordered by peptide then resolution -- ProteinMPNN (no MHC)

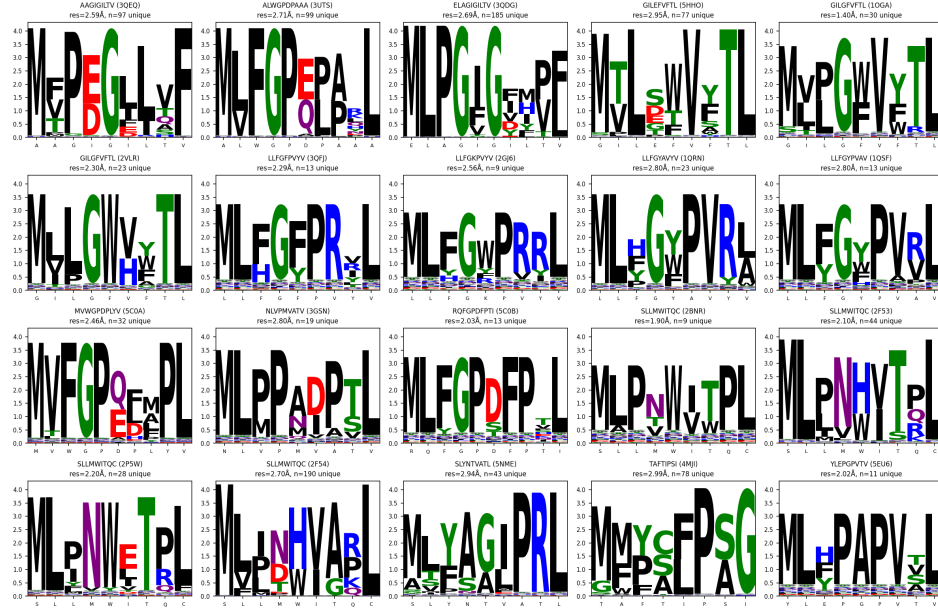


Figure S4b: Per-structure design logos, ProteinMPNN (no MHC).

Per-structure design enrichment logos (bits), ordered by peptide then resolution -- ESM-IF1

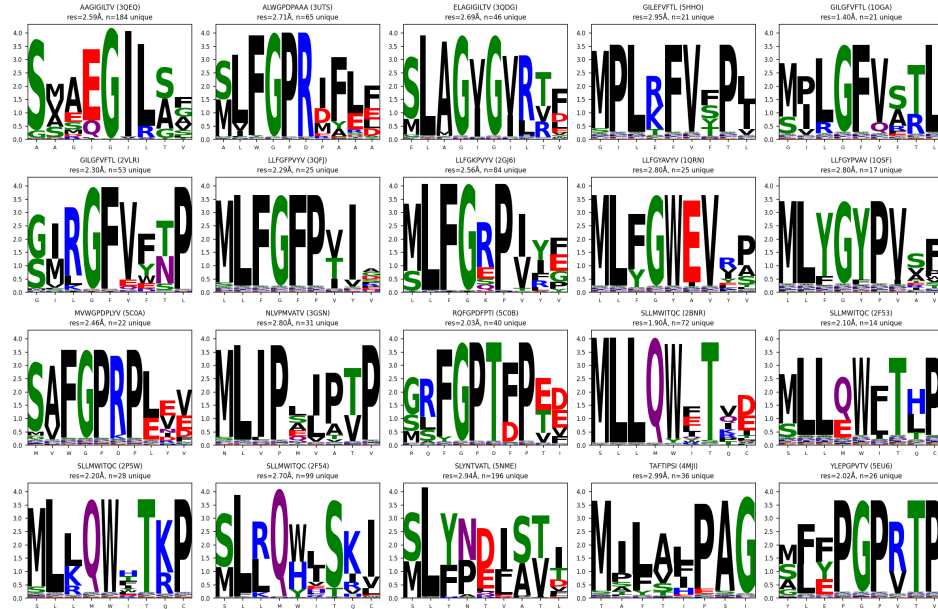


Figure S4c: Per-structure design logos, ESM-IF1.

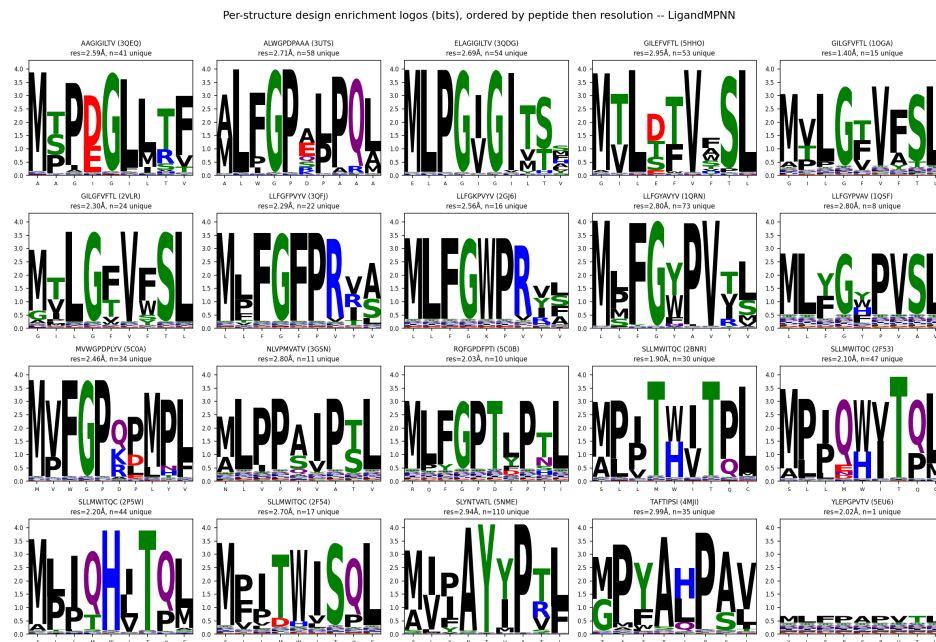


Figure S4d: Per-structure design logos, LigandMPNN.

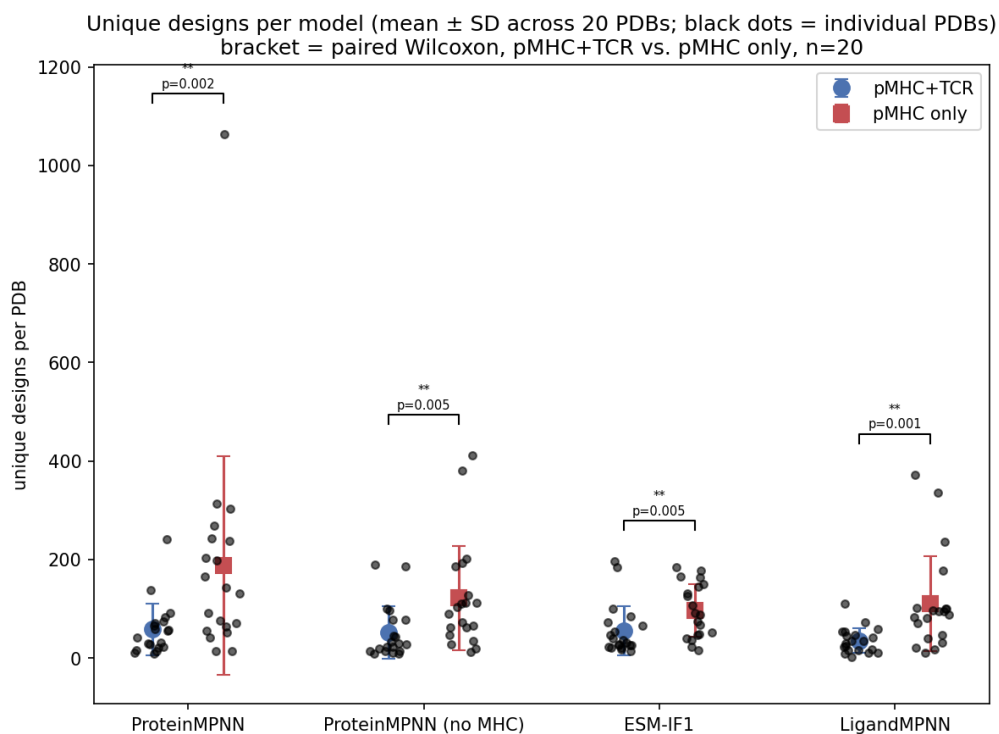


Figure S5: Unique designs per model, mean $\pm$ SD across structures, pMHC+TCR vs. pMHC only.

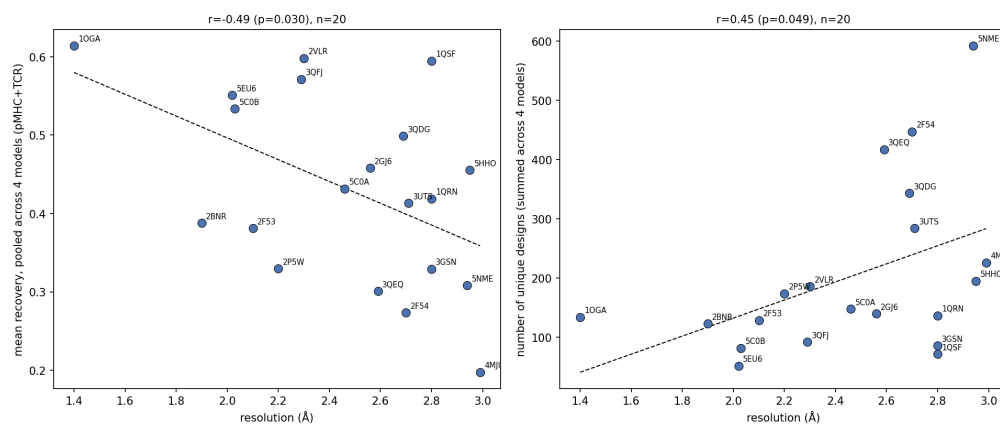


Figure S6: **Resolution and unique-design count vs. mean recovery, pooled across the full 20-structure panel.** $r = -0.49$ ($p = 0.030$) and $r = 0.45$ ($p = 0.049$), $n = 20$.

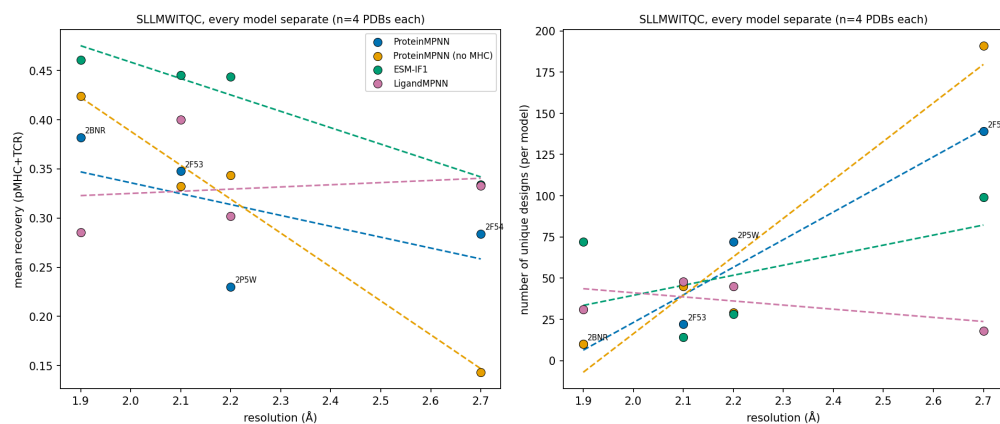


Figure S7: **Resolution dependence in a controlled, same-peptide replicate set.** Four structures (2P5W, 2BNR, 2F53, 2F54), identical native peptide, near-identical TCR clone; every model plotted separately.

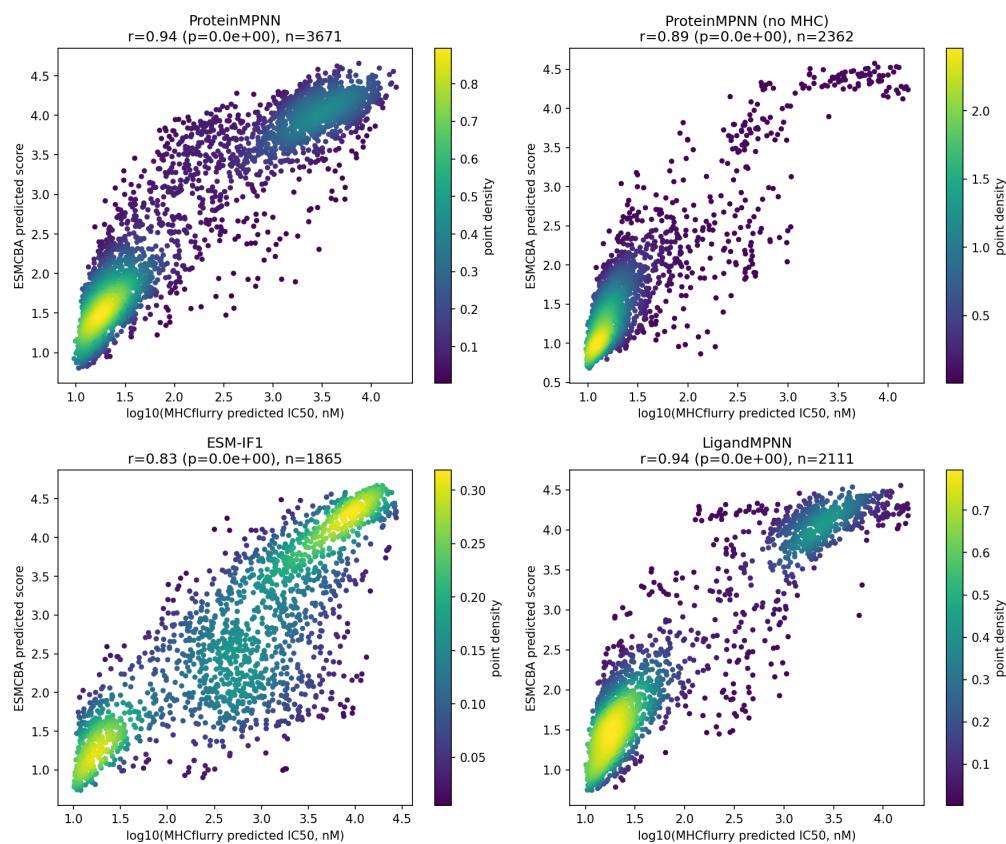


Figure S8a: ESMCBA vs. MHCflurry agreement, pMHC-only designs, by model.

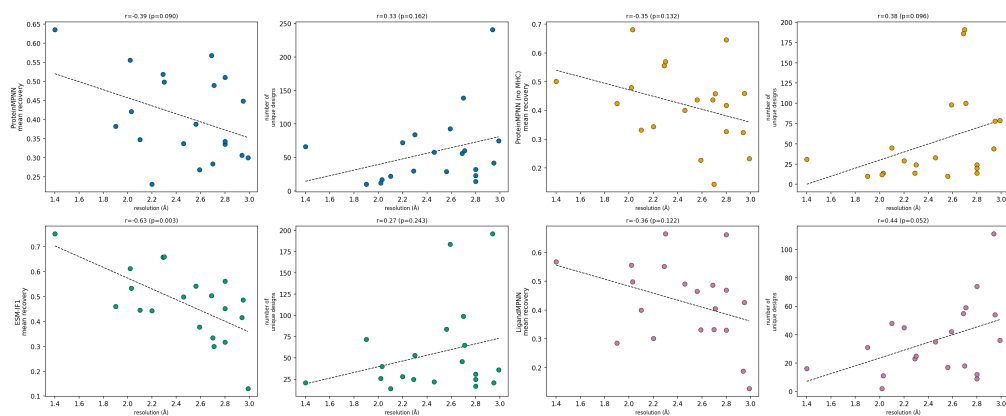


Figure S8b: Per-model resolution/design-count relationship, full panel.

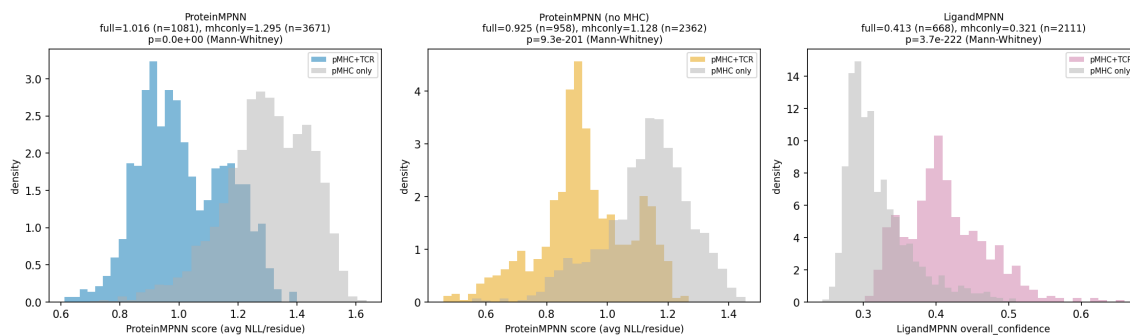


Figure S9a: **Each model's own confidence score for its designs shifts when the TCR is removed**, pooled across all nineteen HLA-A*02:01 structures. ProteinMPNN and ProteinMPNN (no MHC): average per-residue negative log-likelihood, lower is more confident. LigandMPNN: overall confidence, higher is more confident. All three shift toward lower self-assessed confidence without the TCR (Mann-Whitney $p < 10^{-200}$ in all three cases).

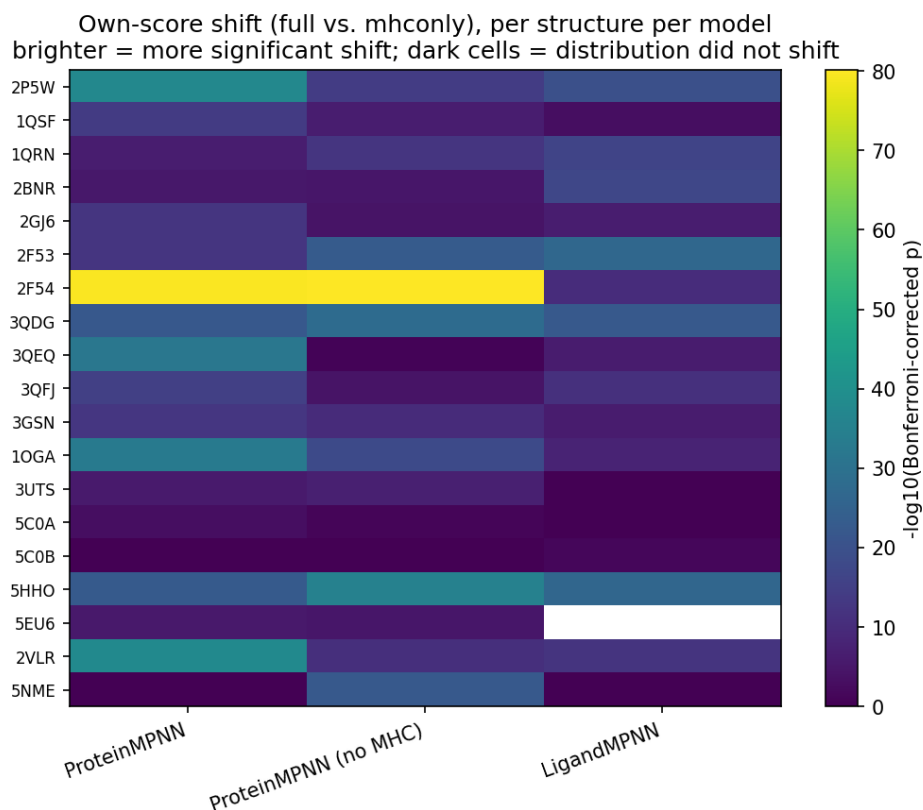


Figure S9b: **The own-score shift is not uniform across the panel.** Bonferroni-corrected significance (full vs. mhonly) per structure per model; dark cells are structures where the model's own-score distribution did not shift.

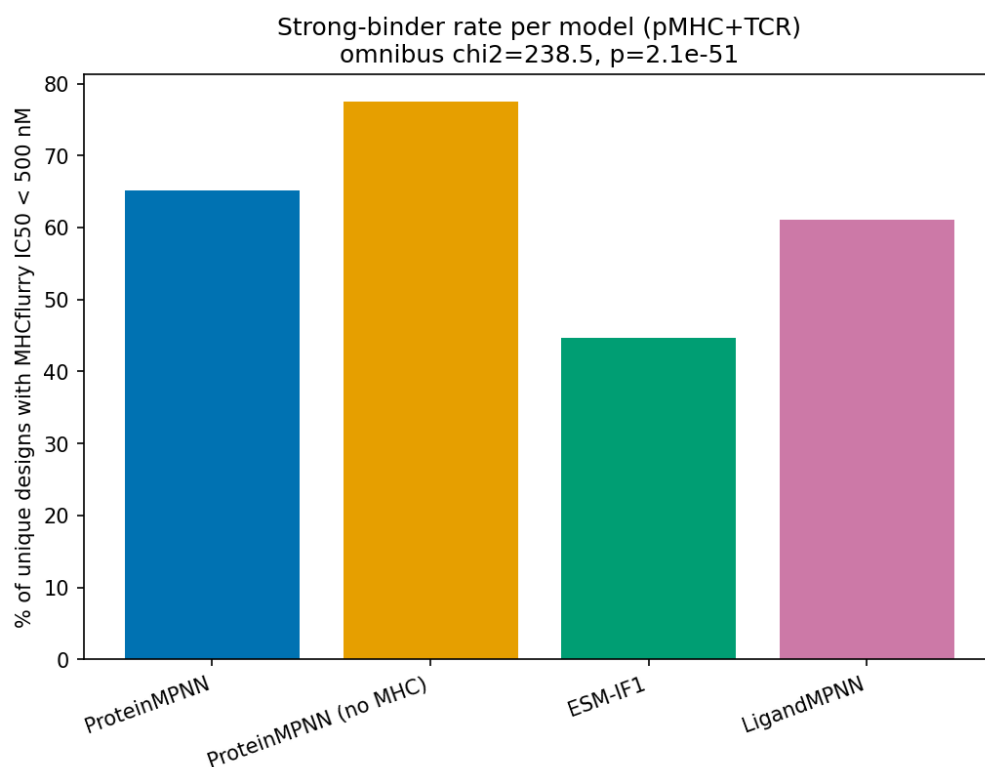


Figure S10: **Removing the TCR raises predicted affinity, mostly for two of the four models.** Strong-binder rate (MHCflurry IC50 < 500 nM), full vs. pMHC-only context, Wilson 95% CI. ProteinMPNN (no MHC) and LigandMPNN both shift up substantially without the TCR; ProteinMPNN and ESM-IF1 barely move. ESM-IF1 is the lowest-affinity model in both conditions.

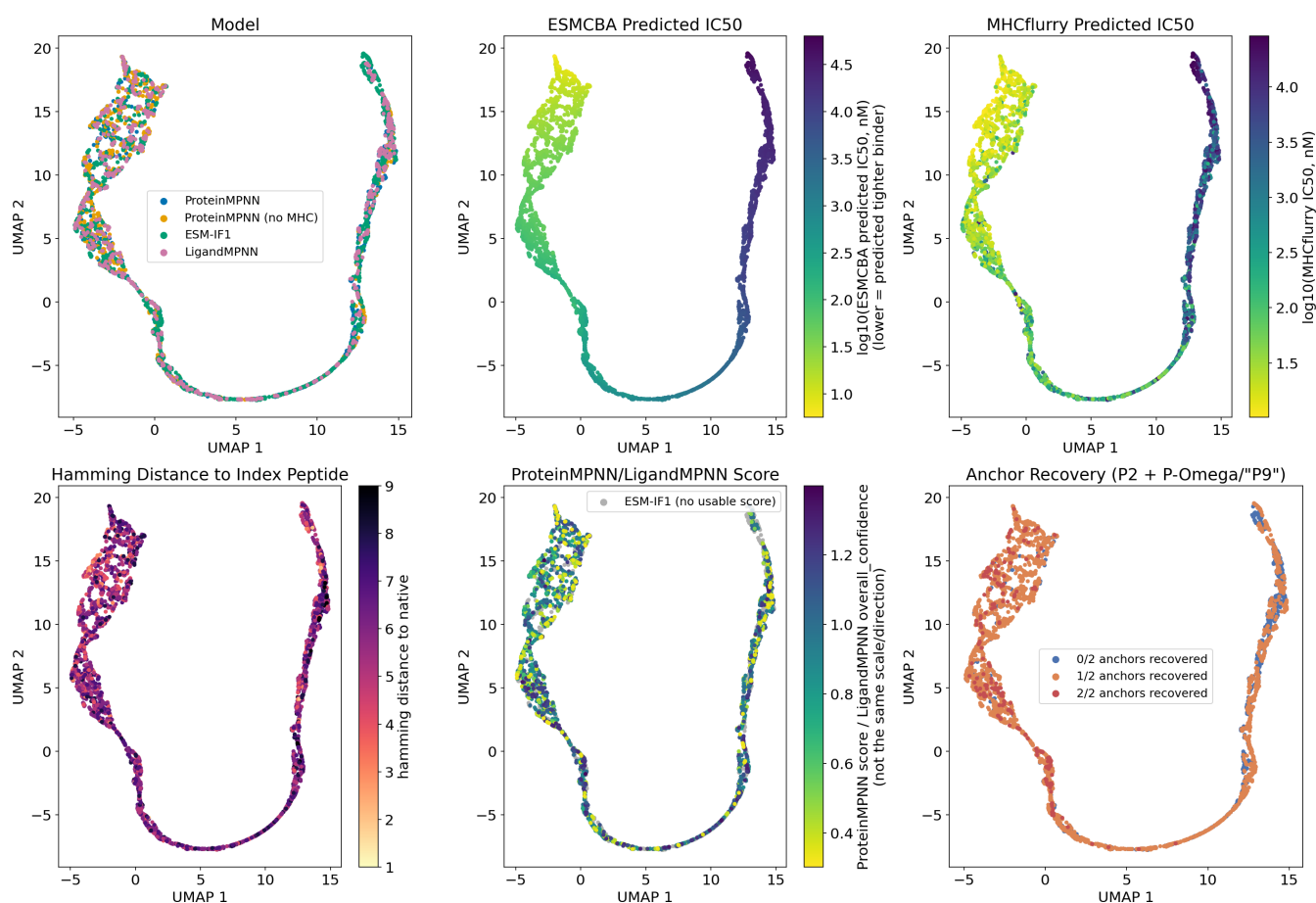


Figure S11: **The sampled sequence space, colored by predicted binding affinity.** One UMAP (ESMCBA embeddings, fit jointly over every unique design), recolored six ways. All four models occupy the same overlapping landscape rather than four separate clusters, and MHCflurry and ESMCBA, built differently and trained differently, trace the identical gradient across it.

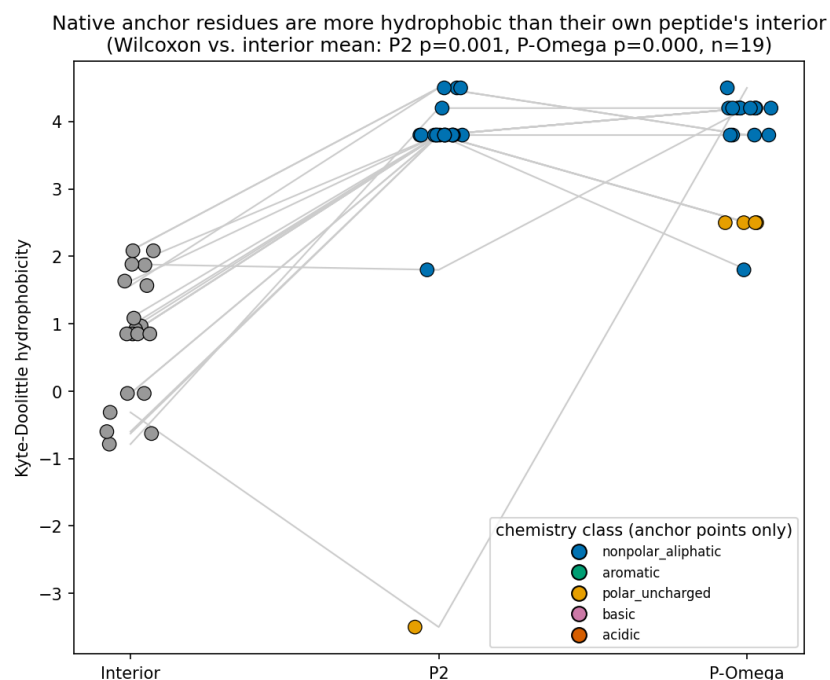


Figure S12: **Both MHC anchors are more hydrophobic than the peptide interior they sit alongside.** Kyte-Doolittle hydrophobicity of the native residue at P2 and P Ω for each of the 19 HLA-A*02:01 structures, plotted against that same peptide's mean over its own interior positions (P3 through the second-to-last, excluding P1). Grey lines connect the three values belonging to one peptide; anchor points are colored by side-chain class. Both anchors sit well above their own interior (paired Wilcoxon, $p = 0.0013$ for P2 and $p = 0.0001$ for P Ω , $n = 19$), so the two pockets do not differ in whether they prefer hydrophobic residues. They differ in how strictly the designs honor that preference, which is the distinction reported in Section 3.2. The four points at P Ω colored as polar are the cysteine-terminated SLLMWITQC/NY-ESO-1 replicates, a residue whose discrete classification understates its hydrophobicity.